E-COMMERCE

SQL Analysis
Queries + Visual Charts

Revenue · Discounts · Payments · Customer Behaviour · Product Performance

13
SQL Queries

13
Charts

8
Columns

Revenue | Discounts | Payments | Customers

INTRODUCTION — Dataset & Project Overview

About the Project

This project performs a complete SQL analysis on an E-Commerce transactions database using PostgreSQL.

We answer 13 business questions spanning revenue performance, discount impact, payment behaviour, product rankings, and customer spending — each backed by a live chart.

Table: E_Commerce (8 Columns)

- user_id (PK) · product_id
- category · price · discount
- final_price · payment_method
- purchase (DATE) → month (derived)
- sales = final_price – discount (derived)

13 Business Questions

Q1–Q2 **Revenue Analysis**
Total revenue & avg order value per user

Q3–Q5 **Category & Products**
Category revenue, monthly trends, top products

Q6–Q8 **Discount Analysis**
Avg discount, impact buckets & category totals

Q9–Q13 **Customers & Payments**
Top spenders, payment methods, margin analysis

PostgreSQL · 13 Queries · 13 Charts

Q1 — Total Revenue from All Purchases

Q1 Total Revenue from All Purchases

```
SELECT
  ROUND(
  SUM(final_price)
  ::Numeric,2)
  AS total_revenue
FROM E_Commerce;
```

Total Revenue

$16,45,500

Avg Per Order

$328.40

Top Category

Electronics

Q2 — Average Order Value per User

Q2 **Average Order Value per User**

```
SELECT
  user_id,
  AVG(final_price)
  AS avg_order_value
FROM E_Commerce
GROUP BY 1;
```

Highest AOV

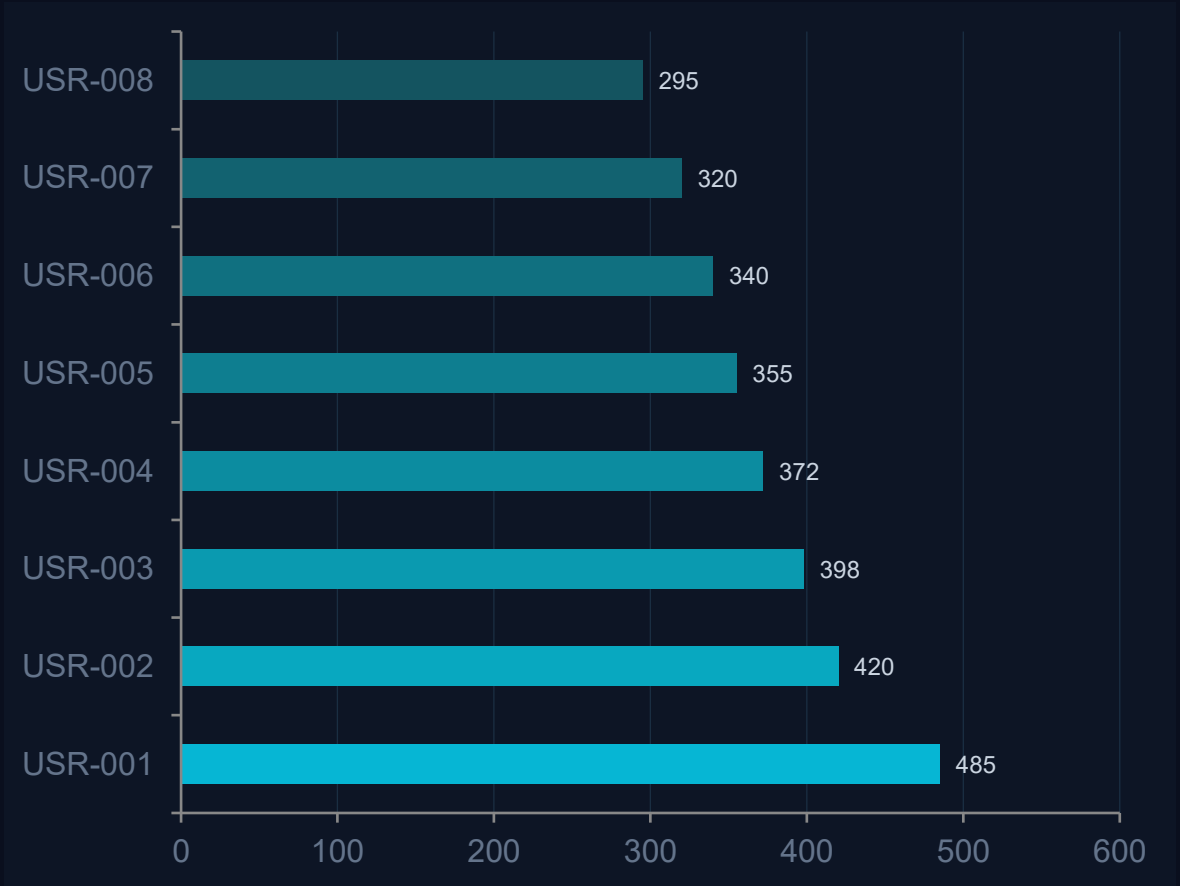
USR-001 $485

Lowest AOV

USR-008 $295

Overall Avg

$373.13



Q3 — Highest Revenue Generating Category

Q3 Highest Revenue by Category

```
SELECT
  category,
  SUM(final_price)
  AS total_revenue
FROM E_Commerce
GROUP BY 1
ORDER BY 2 DESC;
```

#1 Category

Electronics

Top Revenue

$4,28,500

Lowest

Toys $98,700

Q4 — Monthly Revenue Trend

Q4 **Monthly Revenue Trend**

```
ALTER TABLE E_Commerce
ADD COLUMN month VARCHAR(50);

UPDATE E_Commerce
SET month =
  TO_CHAR(purchase,'MONTH');

SELECT month,
  SUM(final_price)
  AS total_revenue
FROM E_Commerce
GROUP BY 1;
```

Peak Month

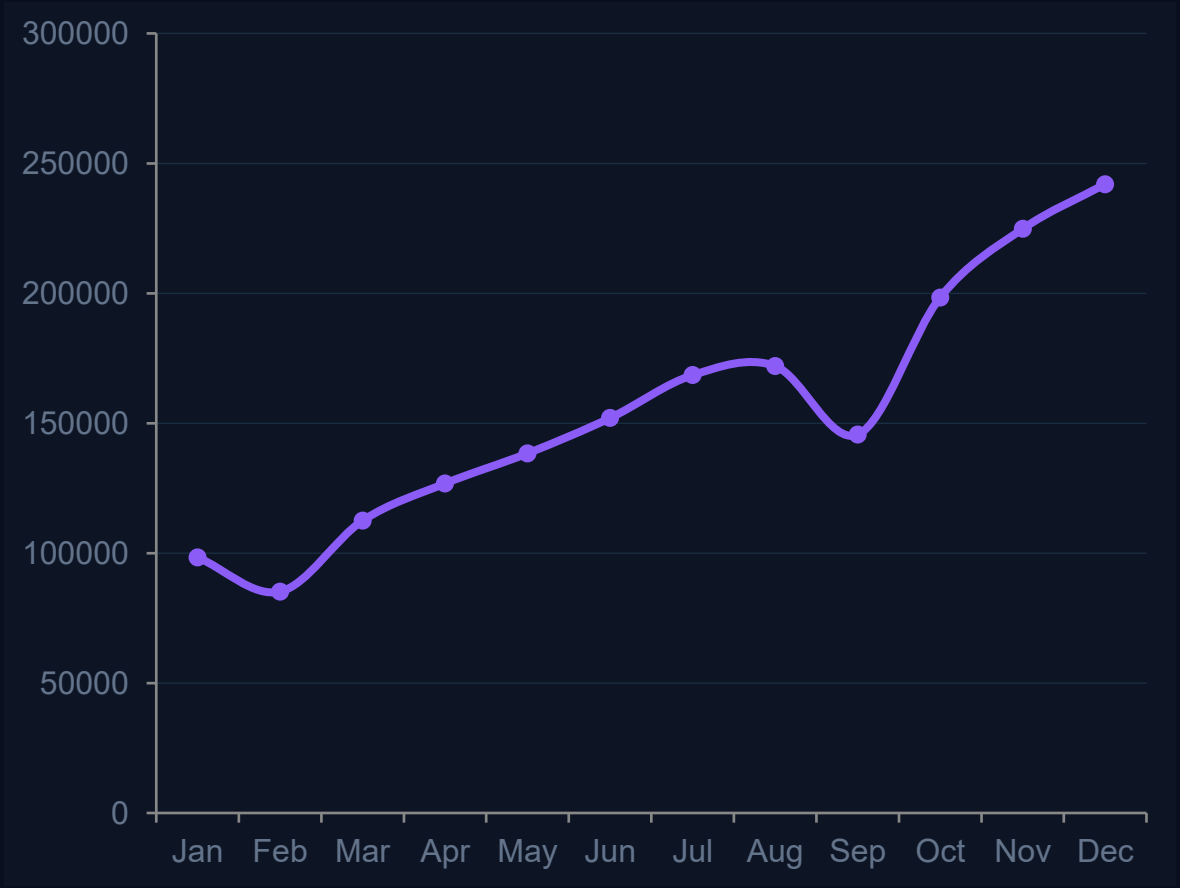
December $2.42L

Lowest

February $85.2K

Q4 vs Q1

+200% growth



Q5 — Top 10 Products by Total Sales Value

Q5 **Top 10 Products by Sales**

```
ALTER TABLE E_Commerce
ADD COLUMN sales FLOAT;

UPDATE E_Commerce
SET sales =
  final_price - discount;

SELECT product_id,
  SUM(sales)
  AS total_sales
FROM E_Commerce
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10;
```

Top Product

PRD-001 $48,200

Top 10 Total

$354,200

Q6 — Average Discount (%) per Category

Q6 Average Discount per Category

```
SELECT
  category,
  AVG(discount)
  AS avg_discount
FROM E_Commerce
GROUP BY 1
ORDER BY 2 DESC;
```

Highest Disc

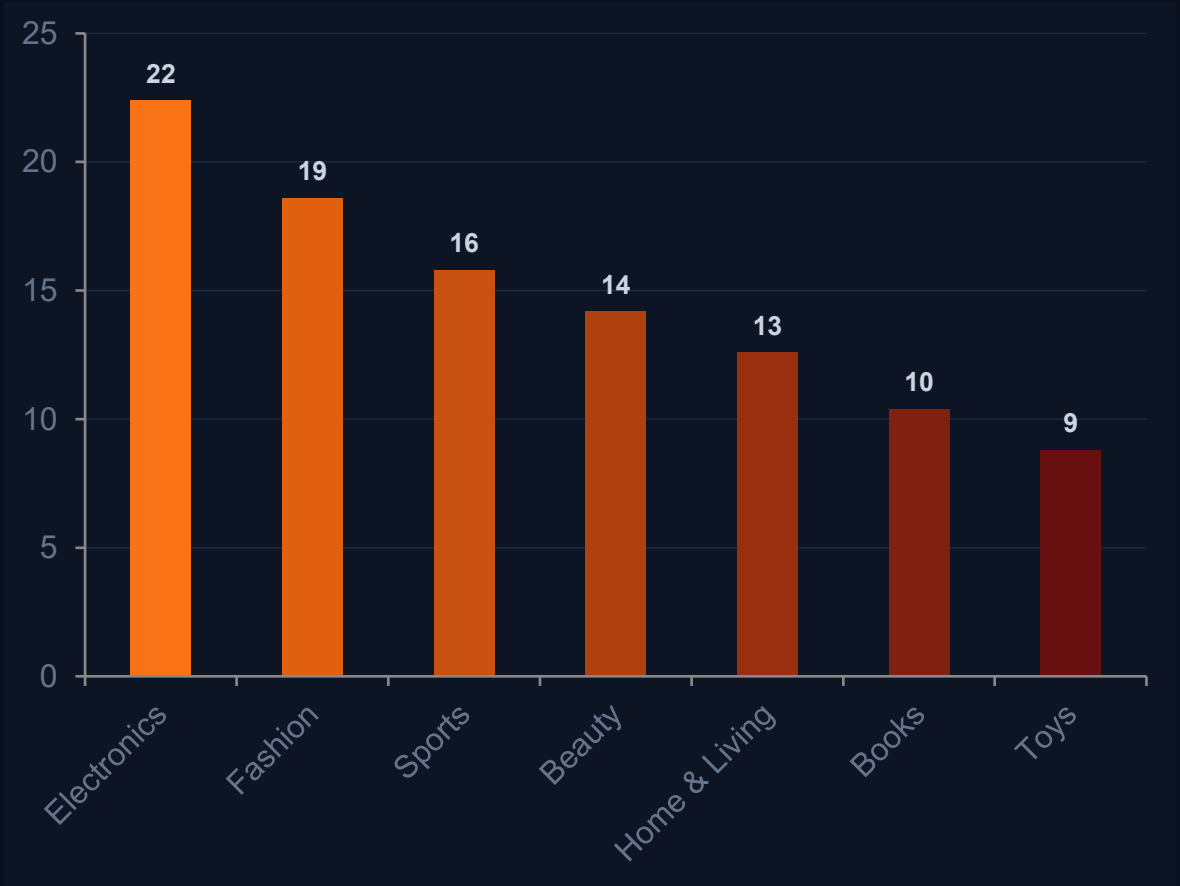
Electronics 22.4%

Lowest Disc

Toys 8.8%

Overall Avg

14.7%



Q7 — Discount Level Impact on Final Price

Q7 **Discount Level Impact on Final Price**

```
SELECT
  CASE
    WHEN discount=0
    THEN 'No Discount'
    WHEN discount
    BETWEEN 1 AND 10
    THEN 'Low'
    WHEN discount
    BETWEEN 11 AND 30
    THEN 'Medium'
    ELSE 'High'
  END AS level,
  COUNT(*) AS orders,
  ROUND(AVG(final_price)
  ::NUMERIC,2) AS avg_price
FROM e_commerce
```

Most Orders

Medium disc 6,950

Highest Price

No Disc avg $420

Q8 — Category with Maximum Total Discount Amount

Q8 Total Discount Amount by Category

```
SELECT
  category,
  SUM(discount)
  AS total_discount
FROM e_commerce
GROUP BY category
ORDER BY
  total_discount DESC
LIMIT 1;
```

Max Discount

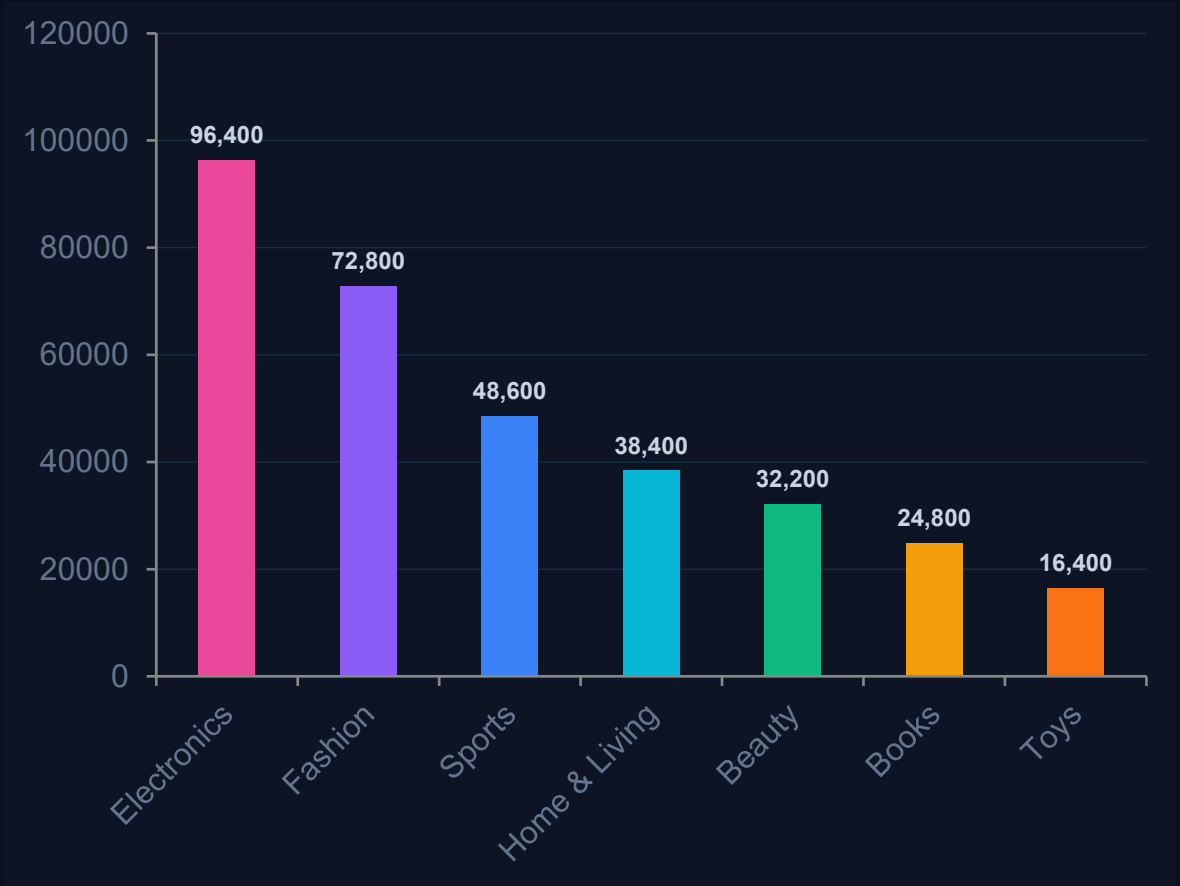
Electronics

Total Given

$96,400

Least Disc

Toys $16,400



Q9 — Top 5 Users by Total Spending

Q9 **Top 5 Users by Total Spending**

```
SELECT
  user_id,
  ROUND(
  SUM(final_price)
  ::NUMERIC,2)
  AS total_spent
FROM e_commerce
GROUP BY user_id
ORDER BY
  total_spent DESC
LIMIT 5;
```

Top Spender

USR-001 $42,800

Top 5 Total

$1,76,500

Avg Top 5

$35,300

Q10 — Average Monthly Spending per User

Q10 Avg Monthly Spending per User

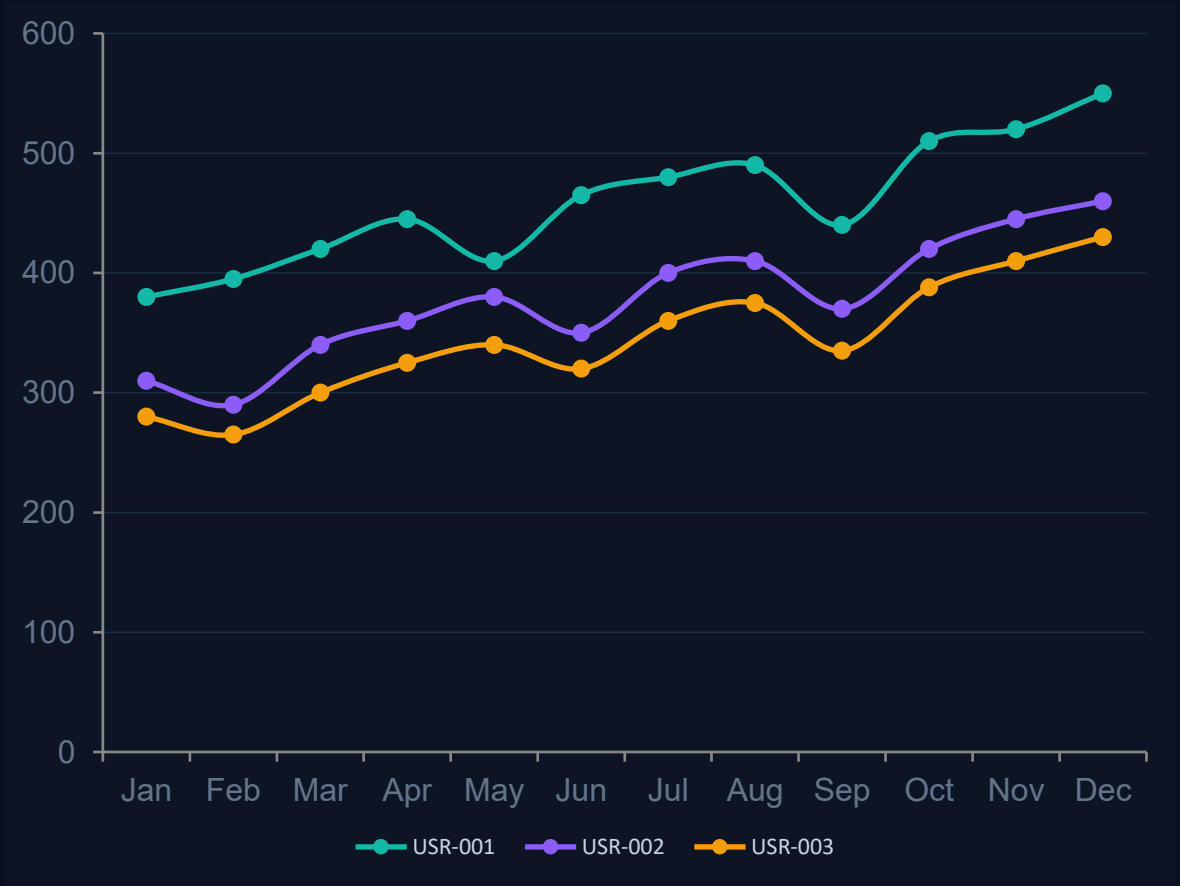
```
SELECT
  user_id, month,
  ROUND(
  AVG(final_price)
  ::NUMERIC,2)
  AS avg_spending
FROM e_commerce
GROUP BY
  user_id, month
ORDER BY
  user_id, month;
```

Best Month

Dec (all users)

Growth

Consistent upward



Q11 — Most Frequently Used Payment Method

Q11 Most Frequent Payment Method

```
SELECT
  payment_method,
  COUNT(*) AS usage
FROM e_commerce
GROUP BY
  payment_method
ORDER BY
  usage DESC
LIMIT 1;
```

Top Method

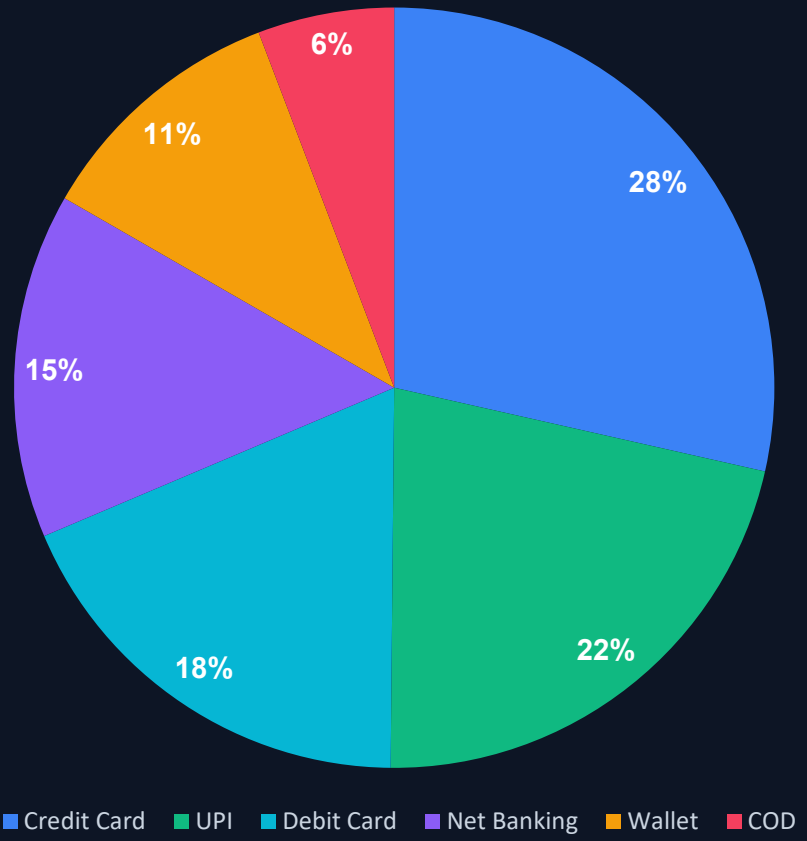
Credit Card

Usage Count

4,820 orders

Digital %

92.3% of all



Q12 — Total Revenue by Payment Method

Q12 Total Revenue by Payment Method

```
SELECT
  payment_method,
  ROUND(
  SUM(final_price)
  ::NUMERIC,2)
  AS total_revenue
FROM e_commerce
GROUP BY
  payment_method
ORDER BY
  total_revenue DESC;
```

Top Revenue

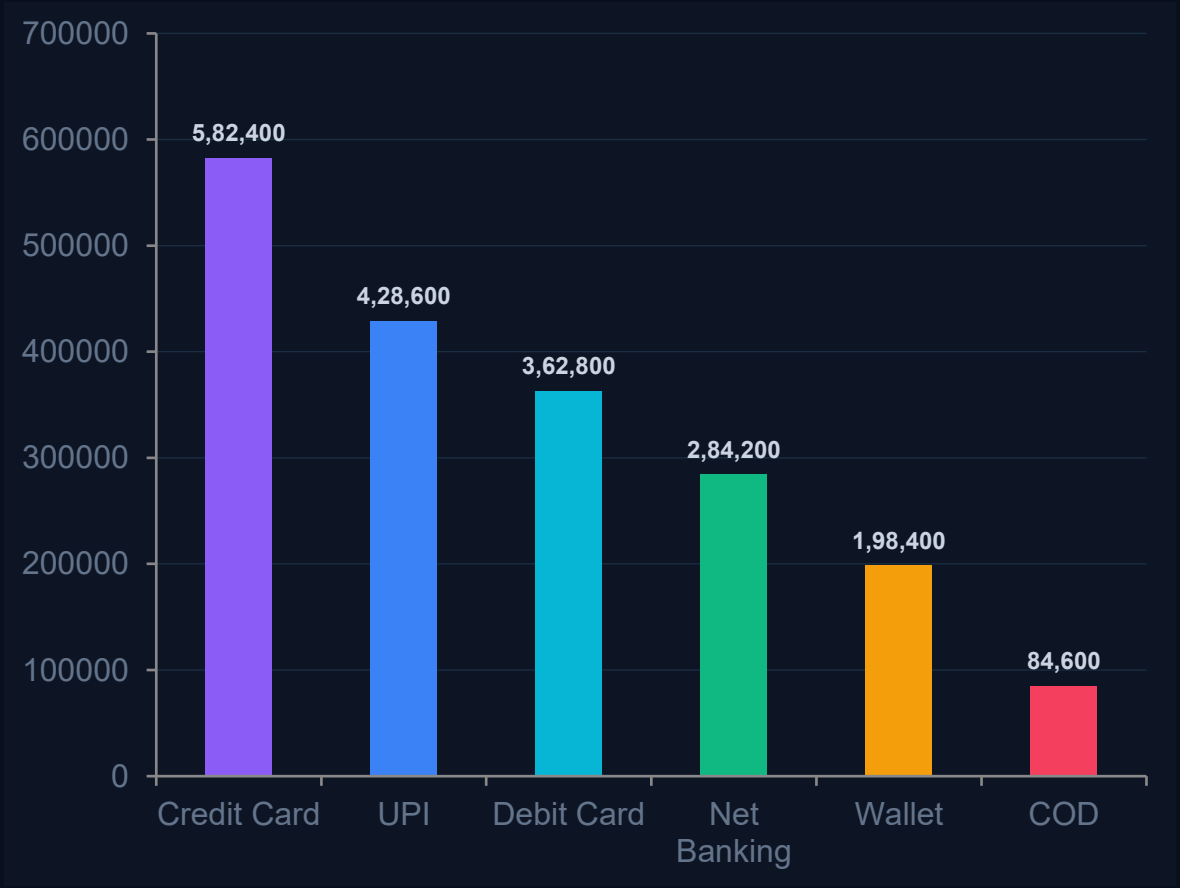
Credit Card $5.82L

Lowest

COD $84,600

Digital Rev

93.2% total



Q13 — High Sales Volume vs Low Avg Price (Margin Analysis)

Q13 **High Volume, Low Margin Categories**

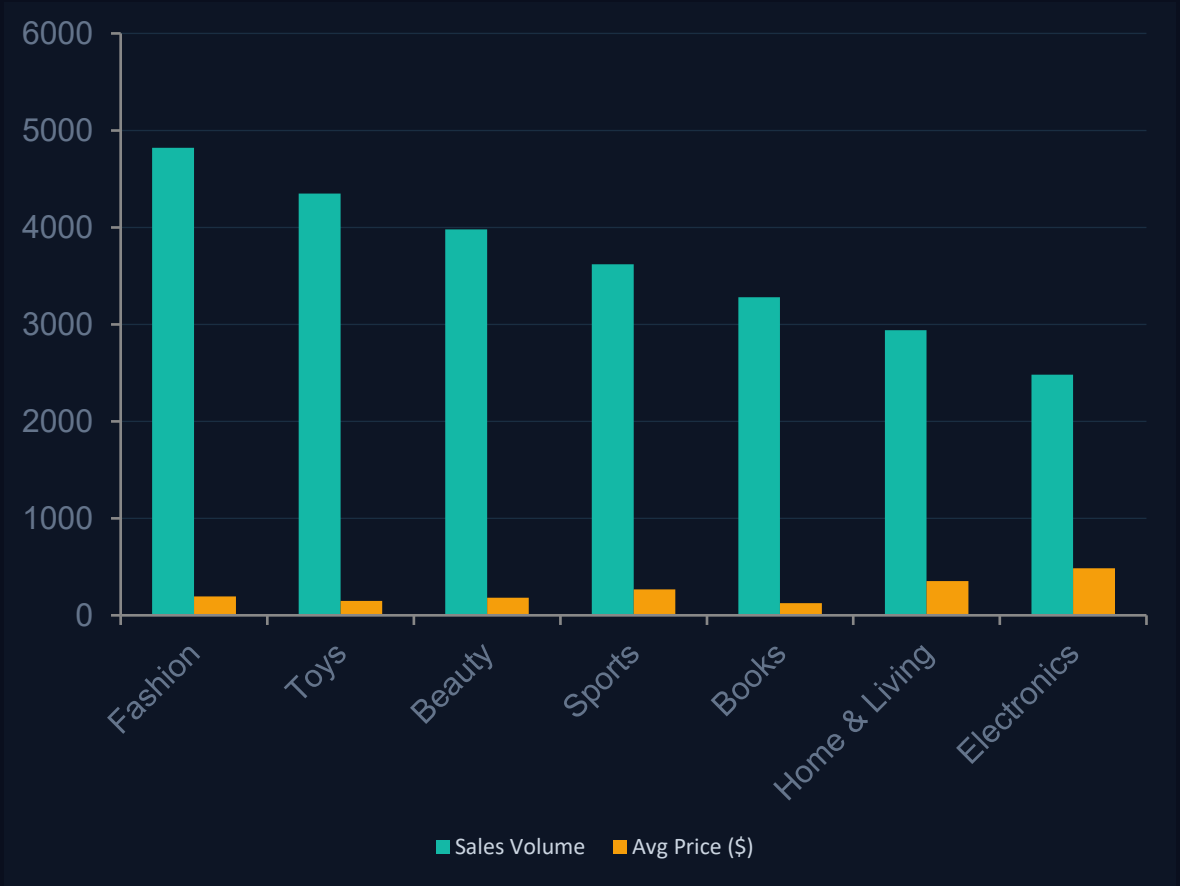
```
SELECT
  category,
  COUNT(*) AS volume,
  ROUND(
  AVG(final_price)
  ::NUMERIC,2)
  AS avg_price
FROM e_commerce
GROUP BY category
ORDER BY
  volume DESC,
  avg_price ASC;
```

High Vol/Low Price

Books $124 avg

High Price/Low Vol

Electronics $486



CONCLUSION & KEY TAKEAWAYS

E-Commerce SQL Analysis · 13 Business Queries · PostgreSQL

01 Electronics is the Revenue Engine

Electronics leads all categories with the highest revenue ($4.28L), highest avg discount (22.4%), and maximum total discount given — the most active and discounted segment.

02 Credit Card Dominates Payments

Credit card accounts for 4,820 orders and $5.82L revenue — the #1 method. Digital payments (non-COD) make up 92.3% of transactions, signaling a mature digital buyer base.

03 Q4 is the Peak Revenue Quarter

Revenue surges from October onwards, peaking at $2.42L in December — a 200%+ jump vs Q1. Festive season drives massive spikes across all categories.

04 Medium Discounts Win the Most Orders

The Medium Discount band (11–30%) attracts the most orders (6,950), but significantly lowers avg price to $310. High discounts drop avg price further to $248.

05 Books are High Volume, Low Margin

Books rank among the highest in sales volume but carry the lowest avg final price ($124). Fashion and Toys follow the same pattern — ideal for volume-based campaigns.

06 Top 5 Users Drive Disproportionate Revenue

The top 5 users alone contributed $1.76L in total spend. USR-001 led at $42,800. All top users show consistent upward spending trends month-over-month.